

Genetic Optimization & Automation of RFIC Passives

Chips & AI Workshop Hackathon

Ryan Wans

September 20, 2026

Principal findings

- The complete geometry-to-network workflow runs with Palace and openEMS. A first HFSS comparison now adds a third solver.
- Refining the field around conductors reduced inductor L disagreement from about 42% to **2.35% at 10 GHz**.
- The compact transformer agrees within **3.14% in self- L** at 10 GHz in the original zero-oxide-loss benchmark.
- The newer lossy benchmark exposes a larger Palace self- L discrepancy. Spatial convergence remains the central validation task.

Source: EXTRACTION_RESULTS.md; TRANSFORMER_RESULTS.md; HFSS benchmark REPORT.md (2026-09-20).

Project scope and goals

Implemented scope

- Parameterized symmetric inductors and interleaved transformers.
- Checked GDS, explicit terminals and versioned stack/material definitions.
- Full multiport EM extraction and differential $L/R/Q/M/k$.
- Local workbench for variations, sweeps, meshes and result comparison.

Design objective

Develop a reproducible path from geometry exploration to defensible passive-device selection.

- Initial inputs specify geometry in μm .
- Target-driven seeds now assist exploration.
- Validated RF optimization remains future work.

Source: README.md; EXPLORATION.md; WEB.md. Modified CMOS5L routing profile, not a generic foundry-qualified PDK.

Progress through September 20

Stage	Evidence delivered	Status
Layout	Deterministic GDS, topology checks, KLayout/KLive	Working
Extraction	Palace/openEMS networks, receipts, plots and reuse	Working
Benchmarks	Inductor mesh study, full four-port transformer	Completed cases
Materials	Versioned profiles, matched frequency-dependent loss	Adapter verified
Exploration	Analytical seeds, bounded sweeps, mesh/cost inspection	Implemented
HFSS	Two-frequency solved archive and CLI comparison	First comparison
Optimization	Reliable objective scoring, DRC gate, automated search	Pending

Source: README.md; TECHNOLOGY_PROGRESS.md; NEXT_PHASE_PROGRESS.md, latest dated entries; HFSS REPORT.md.

Reproducible extraction workflow

- ① **Specify:** geometry parameters and a technology snapshot.
- ② **Generate and check:** GDS, winding connectivity, via coverage and terminals.
- ③ **Build the common physical model:** solids, materials, port sheets and enclosure.
- ④ **Mesh and solve:** independent FEM/FDTD adapters and every excitation.
- ⑤ **Collect and compare:** full S/Z , differential metrics, diagnostics and provenance.

Hashes, frozen workers and completion receipts support result reuse and reject changed or incomplete inputs/results.

Source: README.md; EXTRACTION.md. GDS generation uses a 5 nm source grid and 1 nm database unit.

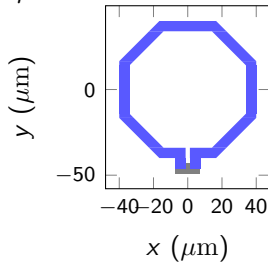
Tool roles in this project

Tool	Role	Project evidence / constraint
RapidPassives ports	Parametric geometry	Adapted generators for modified CMOS5L
gdsfactory + KLayout	GDS assembly, topology, inspection	Deterministic export and independent region checks
Gmsh + Palace	Tetrahedral mesh, frequency-domain FEM	Surrounding-field refinement materially changes L
CSXCAD + openEMS	Cartesian geometry, time-domain FDTD	Fine via cells constrain the time step
HFSS / AEDT	Commercial comparison solver	Two solved frequencies on a licensed host

Source: README.md; PROVENANCE.md; EXTRACTION.md; TRANSFORMER_RESULTS.md; HFSS REPORT.md.

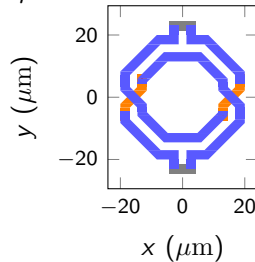
Physical test structures

80 μm one-turn inductor



6 μm trace, 2.5 μm spacing
Two ports, no center tap or shield

40 μm 1 $\times$ 1 transformer



3 μm trace, 2.5 μm spacing
Four ports, four 0.42 μm TopVia1 cuts

Source: Actual GDS: L_compare.gds and archived T_smoke.gds. Blue: TopMetal1; orange: Metal4; red: TopVia1; gray: Metal1. Axes have different scales.

Test-structure coverage

Structure	Ports / samples	Completed evidence
80 μm inductor	2 / 2–20 GHz	Six mesh-study cases, both solvers, full matrices
40 μm transformer	4 / 2–20 GHz	Both solvers, all excitations, zero oxide loss
Lossy 40 μm transformer	4 / 10, 20 GHz	Palace, openEMS and HFSS, oxide $\tan \delta = 0.01$
24 μm material fixture	2 / 10, 20 GHz	Six jobs verify material adapter and aggregation
80 μm transformer	4 / five samples	Palace complete; no completed two-solver comparison
100 μm tapped transformer	6 / prepared	Model and 128,197-tetrahedron mesh; no full solve

Source: EXTRACTION_RESULTS.md; TRANSFORMER_RESULTS.md; material-adapter README.md; HFSS REPORT.md.

Technology selection is more than a layer map

- This project uses a **modified CMOS5L routing profile** for easy PGS support. TopMetal1 connects through TopVia1 directly to Metal4, then through Via3 to Metal3.
- Stock SG13G2 TopVia1 rules expect **Metal5** below TopMetal1. A stock deck or a file named “tech_mod” does not establish compatibility.
- Versioned profiles configure layer/datatype mappings, heights and material laws. Generator topology and lateral feasibility rules remain CMOS5L-specific.
- A new technology needs a reviewed via topology, material stack and process rule deck, in addition to remapped GDS layers.

Source: README.md; TECHNOLOGIES.md; EXPLORATION.md, DRC source audit. Current checks establish topology, not foundry DRC/LVS.

Modeled conductor stack

Role / GDS layer	Bottom z (μm)	Top z (μm)	Conductivity (MS/m)
TopMetal1 / 126:0	5.40	7.40	27.800
TopVia1 / 125:0	4.55	5.40	2.191
Metal4 / 50:0	4.06	4.55	23.190
Via3 / 49:0	3.52	4.06	1.660
Metal3 / 30:0	3.03	3.52	23.190
Metal1 / 8:0	1.04	1.46	21.640

All exposed terminals end on TopMetal1. Local Metal1 reference bars remain in the GDS even when the patterned ground shield is disabled.

Pin labels and EM-marker layers are annotations. Extruding them as conductors would change the physical model.

Source: README.md; EXTRACTION.md, historical benchmark stack. Profiles snapshot the resolved values for each run.

Dielectrics, substrate and enclosure

Region	Default z interval (μm)	ϵ_r	Conductivity (S/m)
Bulk silicon	-753.75 to -3.75	11.9	2
Epitaxial silicon	-3.75 to 0	11.9	5
Oxide	0 to 8.9	4.1	0 in historical runs
Passivation	8.9 to 9.3	6.6	0
Air	9.3 to 109.3	1	0

- Metal replaces overlapping dielectric. Palace fragments volumes into disjoint regions; openEMS applies conductor priorities.
- The default enclosure has six PEC faces and 50 μm lateral padding. Air/substrate extents are configurable physical assumptions.
- Boundary distance, substrate thickness, reference wiring and dielectric loss require separate sensitivity studies.

Source: EXTRACTION.md; TECHNOLOGIES.md. Table describes the historical default, not every fixture's resolved domain.

Material dispersion uses frequency-point equivalents

- Immutable profiles store constant, loss-tangent, Debye or Lorentz response laws and a SHA-256 identity.
- At each sample, both solvers receive the same evaluated ϵ' and equivalent conductivity:

$$\sigma_{\text{eq}}(\omega) = \omega\epsilon_0\epsilon''(\omega).$$

- For oxide $\epsilon_r = 4.1$, $\tan \delta = 0.01$: $\sigma = 0.02281$ S/m at 10 GHz and 0.04562 S/m at 20 GHz. HFSS uses zero separate loss tangent to avoid double counting.
- Cost scales as N Palace and $N \times P$ openEMS jobs. This is a frequency-point adapter, not native broadband dispersive FDTD.

Source: TECHNOLOGIES.md; HFSS REPORT.md. No process-fitted Debye/Lorentz poles claimed. Historical zero-loss archives remain unchanged.

Ports and differential quantities define the measured DUT

Terminal sheets span Metal1 top ($z = 1.46 \mu\text{m}$) to TopMetal1 bottom ($z = 5.4 \mu\text{m}$). They use local reference bars, 50Ω impedances and no de-embedding.

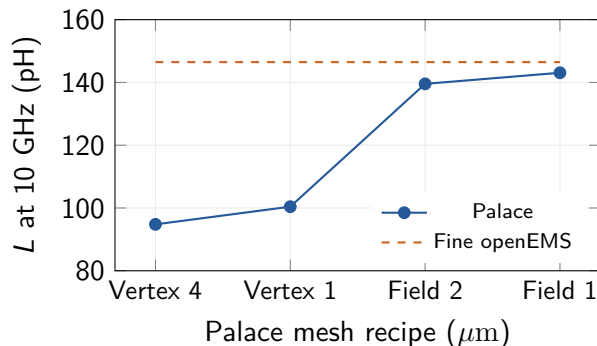
$$Z = 50(I + S)(I - S)^{-1}, \quad Z_d = T^T Z T$$

$$R_i = \Re Z_{d,ii}, \quad L_i = \frac{\Im Z_{d,ii}}{2\pi f}, \quad Q_i = \frac{\Im Z_{d,ii}}{\Re Z_{d,ii}}$$

- Every S column comes from a physical excitation with other ports matched.
- T applies equal/opposite winding currents. Unused currents, including center taps, are zero in differential Z .
- Signed M and $k = M/\sqrt{L_1 L_2}$ follow terminal order. Reversing one winding reverses their signs.
- Moving a port sheet or changing ground/reference wiring changes the DUT.

Source: EXTRACTION.md; TRANSFORMERS.md; HFSS REPORT.md. No separately validated transformer dot convention.

Mesh placement matters more than nominal size alone



The surrounding field needs resolution

Vertex-only refinement left a large discrepancy even at $1\ \mu\text{m}$.

Distance-based volume refinement around conductor surfaces brought L closer to openEMS.

The last refinement still changes L and R . Close agreement alone does not establish convergence.

Source: EXTRACTION_RESULTS.md, September 13. Historical vertex-only cases precede the corrected field-refinement recipe.

FEM and FDTD have different meshing sensitivities

Palace / Gmsh

- Conforming tetrahedra partition conductor and dielectric volumes.
- Near-conductor size, minimum size and grading distance control field resolution.
- Basis order and linear tolerance are separate from spatial resolution.
- Field-error indicators are not calibrated percentage errors in L or Q .

openEMS

- Cartesian grid anchors preserve material interfaces, ports and every via.
- Tiny via cells constrain the time step, even with coarse nominal xy spacing.
- Residual-energy decay verifies time termination, not spatial accuracy.
- Missing/unused vias invalidate a run. Grid budgets guard memory cost.

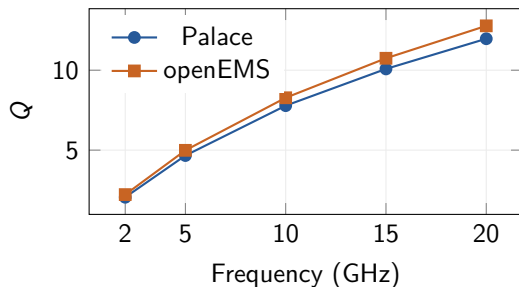
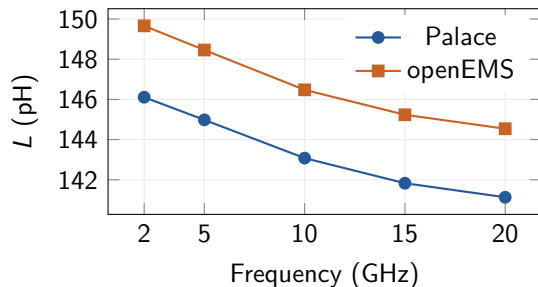
Source: EXTRACTION.md; TRANSFORMER_RESULTS.md; EXPLORATION.md. Equal nominal mesh sizes are not equivalent discretizations.

Thin conductors and vias need explicit attention

- The transformer fixture has four $0.42\text{ }\mu\text{m}$ via cuts. The corrected openEMS grid samples via edges and centers; the earlier larger-device grid missed cuts.
- Its 236,196 cells require a $3.64 \times 10^{-16}\text{ s}$ time step and roughly 572,000–586,000 steps per excitation to reach -50 dB .
- Finite-conductivity volume models must resolve current crowding and skin/proximity effects. For TopMetal1, $\delta \approx 0.95\text{ }\mu\text{m}$ at 10 GHz and $0.67\text{ }\mu\text{m}$ at 20 GHz.
- Nominal in-plane spacing alone cannot certify thin-metal or via accuracy. Inspect actual local cells and extracted R/Q stability.

Source: TRANSFORMER_RESULTS.md; EXPLORATION.md. Skin depth calculated as $\delta = (\pi f \mu_0 \sigma)^{-1/2}$ with $\sigma = 27.8\text{ MS/m}$ and $\mu_r = 1$.

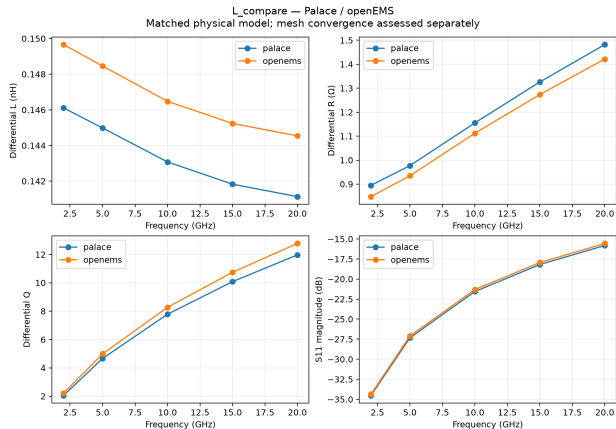
Finest tested inductor comparison



	10 GHz	Palace	openEMS	Symmetric difference
L (nH)		0.143078	0.146476	2.35%
R (Ω)		1.155257	1.112162	3.80%
Q		7.78168	8.27518	6.15%

Source: inductor-2026-09-13/comparison.csv. Difference = $200|a - b|/(|a| + |b|)\%$. Lines connect only the five sampled frequencies.

Inductor comparison: original archived plots



Zero oxide loss; finest tested meshes. Spatial convergence remains unestablished.

Source: docs/benchmarks/inductor-2026-09-13/comparison.png

Mesh cost increases sharply with resolution

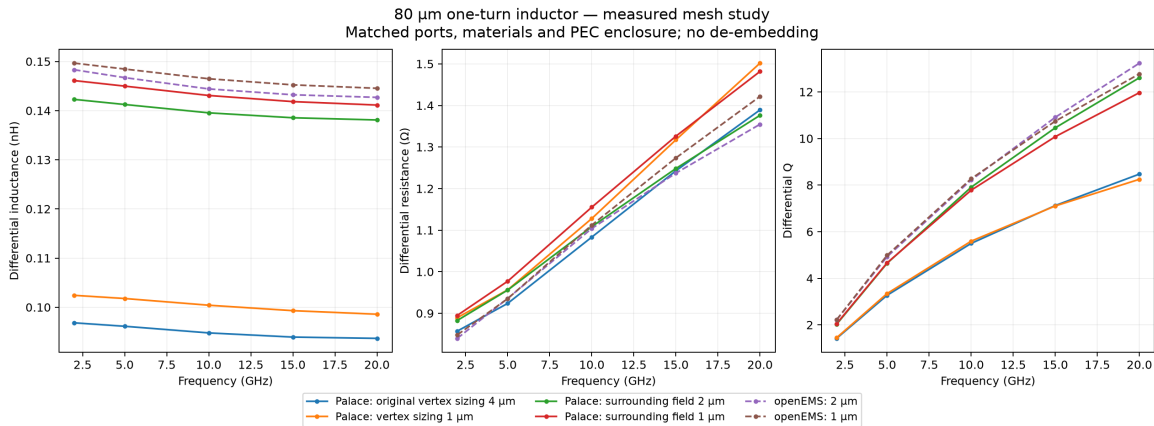
Solver / mesh recipe	Elements or cells	L at 10 GHz (pH)	Solve (s)
Palace, vertex 4 μm	8,998	94.782	8.72
Palace, vertex 1 μm	41,252	100.414	93.31
Palace, field 2 μm	63,402	139.545	382.86
Palace, field 1 μm	149,761	143.078	960.31
openEMS, xy 2 μm	348,160	144.409	225.21
openEMS, xy 1 μm	1,746,030	146.476	3,324.82

Palace uses tetrahedra; openEMS uses Cartesian cells. Counts do not imply equivalent accuracy.

Timing scope: all excitations, four requested CPU threads. Excludes meshing/startup. Some jobs competed for resources, so these observations do not support a controlled speed ranking.

Source: EXTRACTION_RESULTS.md. openEMS stack steps are at most 0.5 / 0.25 μm for the two cases.

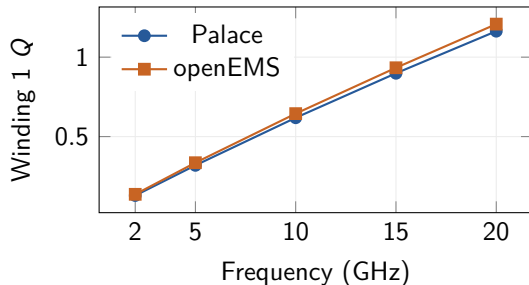
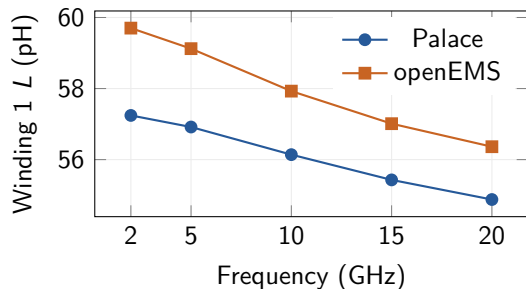
Inductor mesh study: all six cases



Matched physical model; historical vertex-only and surrounding-field mesh recipes.

Source: docs/benchmarks/inductor-2026-09-13/mesh-study.png

Compact transformer agreement across five samples



Across both windings and 2–20 GHz: self- L differs 2.67–4.21%, R 0.95–1.52%, and Q 3.72–5.72%.
Absolute k difference is 0.0153–0.0174.

Sparse vias keep the fixture manageable but produce high resistance and low Q . This is a functional test structure, not an optimized design.

Source: [transformer-2026-09-14/comparison.csv](#); TRANSFORMER_RESULTS.md. Zero oxide loss. Symmetric percentage metric.

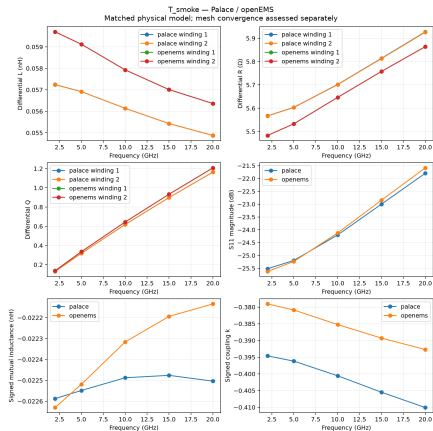
Transformer benchmark at 10 GHz

Quantity	Palace	openEMS	Difference
L_1 (pH)	56.1400	57.9279	3.13%
R_1 (Ω)	5.70202	5.64648	0.98%
Q_1	0.618620	0.644600	4.11%
Signed M (pH)	-22.4878	-22.3162	0.77%
Signed k	-0.400566	-0.385241	0.015325 absolute

- Palace: 36,744 tetrahedra, 239.38 s for 20 frequency/excitation solves.
- openEMS: 236,196 cells, 1,315.25 s summed over four excitations.
- All four excitations reached below -50 dB residual energy. No unused via primitives were reported.

Source: TRANSFORMER_RESULTS.md. Times are not a controlled speed comparison. No completed transformer spatial refinement study.

Transformer comparison: original archived plots



Zero oxide loss; both windings, reflection, signed mutual inductance and coupling.

Source: docs/benchmarks/transformer-2026-09-14/comparison.png

HFSS comparison for the lossy transformer

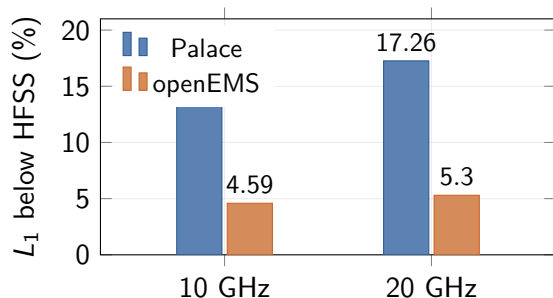
GHz	Solver	L_1 (pH)	R_1 (Ω)	Q_1	M (pH)	k
10	HFSS	60.692	5.7843	0.6593	-22.858	-0.3765
10	Palace	50.416	5.6061	0.5651	-22.325	-0.4431
10	openEMS	57.907	5.6466	0.6444	-22.308	-0.3852
20	HFSS	59.528	6.0256	1.2415	-22.848	-0.3838
20	Palace	49.254	5.8062	1.0660	-22.224	-0.4515
20	openEMS	56.371	5.8639	1.2080	-22.137	-0.3927

Different benchmark conditions: oxide $\tan \delta = 0.01$ and a coarser 15,007-tetrahedron Palace mesh. The original zero-loss comparison used 36,744 tetrahedra.

The change from the older result cannot be attributed to dielectric loss alone.

Source: hfss-transformer-2026-09-20/comparison.csv and REPORT.md; TRANSFORMER_RESULTS.md.

The new discrepancy is strongest in self-inductance



R_1 differs from HFSS by roughly 2–4%; mutual- L magnitude by roughly 2–3%.

Palace's larger $|k|$ accompanies smaller self- L :

$$k = \frac{M}{\sqrt{L_1 L_2}}.$$

Interpretation: investigate self-impedance and field resolution. The cause remains unresolved. HFSS is not yet a validated accuracy reference.

Source: HFSS REPORT.md. These percentages use HFSS as denominator, unlike the earlier symmetric metric.

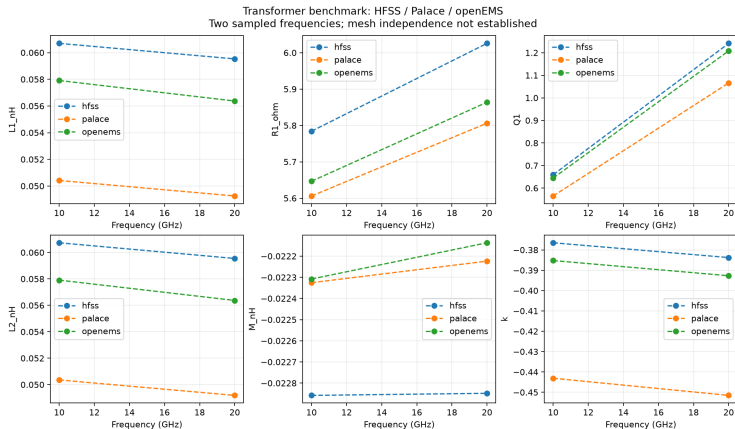
HFSS completion and convergence evidence

GHz	Passes	Tetrahedra	Final ΔS	Solve (s)	Build + solve (s)
10	3	32,720	0.000768	38	477.3
20	3	32,697	0.001409	40	487.2

- Both runs report normal completion and convergence against $\Delta S \leq 0.01$, with two consecutive qualifying passes.
- Review checked identities, port mapping, 50 Ω normalization, oxide loss and saved port positions.
- Tighter adaptation and direct $L/R/Q/k$ stability checks remain necessary. The audit did not reconstruct every solid/contact.
- Two frequencies cannot establish self-resonance, peak Q or bandwidth.

Source: HFSS REPORT.md; project-audit.json; convergence and profile exports.

Lossy transformer: three-solver comparison



Oxide loss included; only 10 and 20 GHz sampled. Mesh independence is not established.

Source: docs/benchmarks/hfss-transformer-2026-09-20/comparison.png

Numerical studies must hold physical assumptions fixed

Numerical controls

- Conductor/field mesh size and grading
- Stack z step and via/port anchors
- FEM basis order and linear tolerance
- FDTD decay threshold and step budget

Physical controls

- GDS geometry and metal thickness
- Material loss and dispersion
- Port sheets and reference conductors
- PEC margin, air and substrate extents

A mesh study keeps physical conditions and frequency samples fixed. Changing the enclosure or loss profile creates a separate physical study.

Use at least three resolutions and assess L , R/Q , M/k independently. A small two-mesh change is sensitivity evidence, not proof of convergence.

Source: EXPLORATION.md; EXTRACTION.md. Acceptance tolerances, including absolute thresholds near zero, remain to be defined.

Validation has several distinct levels

Level	Evidence	Remaining boundary
Layout topology	Via coverage, winding nets, terminal landing, deterministic GDS	Full process DRC/LVS
Software contracts	Port order, S/Z formulas, material partition, stale-result rejection	Physical accuracy
Solver completion	Linear solves, FDTD decay, HFSS adaptive stopping	Spatial convergence
Network sanity	Reciprocity, sampled passivity, positive R , independent parsing	Broadband guarantees
Model validation	Cross-solver comparison and initial mesh sensitivity	Domain/port studies and measurement correlation

Source: tests/; EXTRACTION_RESULTS.md; TRANSFORMER_RESULTS.md; HFSS REPORT.md.

Software test structure

Test family	Representative checks
Layout pipeline	Turn/tap combinations, GDS round trip, missing vias and deliberate winding shorts
Extraction and materials	2/3/4/6-terminal handoff, material volumes, port areas, passive material laws
Networks and receipts	Touchstone order, differential conventions, incomplete solves, hash mismatch and stale reuse
Workbench / exploration	Immutable revisions, queue/cancel/retry, sweep lineage, seed generation and mesh budget guards
HFSS review	Frequency/terminal alignment, provenance rejection and completed-export ingestion

Latest documented full-suite checkpoint: **97 passed**, before later HFSS additions. Subsequent focused checks include discovery/transport and six HFSS review tests.

Source: tests/test_*.py; NEXT_PHASE_PROGRESS.md. Historical test record; no fresh full-suite claim in this presentation.

Exploration tools support controlled studies

- **Starting points:** Mohan-based inductor seeds from target L and geometry bounds. Uncalibrated for these symmetric crossovers.
- **Sweeps:** up to 64 candidates and six axes, immutable plans and a serial queue. Mesh preparation can precede solving.
- **Assessment:** recorded cost and response changes under matching physics. Mesh counts and skin-depth guidance aid inspection.
- **Known gaps:** Q/SRF indicators are budgets rather than validated RF predictions. No transformer coupling surrogate or automated optimizer yet.

Source: EXPLORATION.md; NEXT_PHASE_PROGRESS.md. HFSS workbench import/overlay controls remain pending; CLI review is available.

Priorities before automated design selection

- 1 **Resolve numerical sensitivity.** Tighten HFSS adaptation; refine Palace fields/ports/vias and openEMS cells under identical physics. Track $L/R/Q/M/k$ changes.
- 2 **Define acceptance criteria.** Agree metric tolerances and test enclosure, substrate, reference and port sensitivity separately.
- 3 **Establish process checks.** Validate a modified-CMOS5L DRC deck, including via enclosures, spacing, slotting and fill. Include final fill geometry in EM.
- 4 **Connect design objectives.** Broaden frequency coverage, calibrate seed/scoring models and add budget/cache/resume controls before optimization.

Source: HFSS REPORT.md, Next validation; EXPLORATION.md. Proposed next work.

Evidence map and reproducibility

Paths below are relative to the repository root.

Topic	Primary project source
Scope / conventions	README.md, docs/EXTRACTION.md
Inductor study	docs/EXTRACTION_RESULTS.md docs/benchmarks/inductor-2026-09-13/
Transformer study	docs/TRANSFORMER_RESULTS.md docs/benchmarks/transformer-2026-09-14/
HFSS comparison	docs/benchmarks/hfss-transformer-2026-09-20/
Materials / exploration	docs/TECHNOLOGIES.md, docs/EXPLORATION.md
Validation history	docs/NEXT_PHASE_PROGRESS.md, tests/

Archives retain numerical networks, raw exports and provenance. This deck uses existing evidence and does not launch new EM solves.

Thank you!

Questions?

Contact: `wansr@purdue.edu` or `contact@ryanwans.com`